

$$1) f(x, y) = \sqrt{xy}$$

$$\{(x, y) \in \mathbb{R}^2 \mid xy \geq 0\} =$$

$$(-\infty, 0] \times (-\infty, 0] \cup [0, \infty) \times [0, \infty)$$

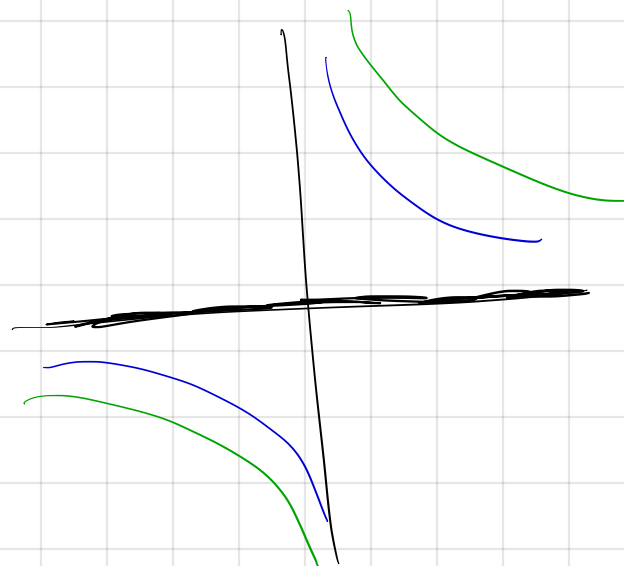
$$\sqrt{xy} = c$$

$$y = \frac{c^2}{x}$$

$$- y = 0 \quad c = 0$$

$$- y = \frac{1}{x} \quad c = 1$$

$$- y = \frac{2^2}{x} \quad c = 2$$



$$2) \lim_{(x, y) \rightarrow (1, 1)} \frac{\sqrt{xy}}{x^2 + y^2} = \frac{\sqrt{1 \cdot 1}}{1^2 + 1^2} = \boxed{\frac{1}{2}}$$

1) Evaluate $f(a, b)$

Check for $\frac{0}{0}, \frac{\infty}{\infty}, 0 \times \infty, 1^\infty, \infty - \infty, 0^0, \infty^0$

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b)$$

2) Plug in direction, e.g. $y = g(x)$, $x = h(y)$

$$\lim_{x \rightarrow a} f(x, g(x)) \text{ OR } \lim_{y \rightarrow b} f(h(y), y)$$

3) Compare limits. If not equal

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = \text{DNE}$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - 2y^2}{x^2 + y^2}$$
$$\frac{0^2 - 2 \cdot 0^2}{0^2 + 0^2} = \frac{0}{0}$$

positive x-axis: $y = 0$
positive y-axis: $x = 0$

$$\lim_{x \rightarrow 0^+} \frac{x^2 - 2 \cdot 0^2}{x^2 + 0^2} =$$

$$\lim_{x \rightarrow 0^+} 1 = 1$$

$$\lim_{y \rightarrow 0^+} \frac{0^2 - 2y^2}{0^2 + y^2} =$$

$$\lim_{y \rightarrow 0^+} -2 = -2$$

$\neq$
DNE

$$\lim_{(x,y) \rightarrow (0,0)} \frac{2xy}{x^3 + yx}$$

$$\textcircled{1} y = mx$$

$$\lim_{x \rightarrow 0} \frac{2x(mx)}{x^3 + (mx)x}$$

$$= \lim_{x \rightarrow 0} \frac{x^2 \cdot 2m}{x^2(x+m)}$$

$$= \lim_{x \rightarrow 0} \frac{2m}{x+m} = \frac{2m}{0+m} = 2$$

$$\textcircled{2} y = x^2$$

Partial derivatives: $f(x, y)$

$$\frac{\partial}{\partial x} f(x, y) = \frac{\partial f}{\partial x}(x, y) = f_x(x, y)$$
$$\frac{\partial f}{\partial x} = f_x$$

$$\frac{\partial^2}{\partial x^2} f(x, y) = \frac{\partial^2 f}{\partial x^2} = f_{xx}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = f_{yx} = (f_y)_x$$

$$3) f(x, y) = (xy)^{3/2} - (xy)^{2/3}$$

$$\frac{\partial f}{\partial x} = \frac{3}{2}(xy)^{1/2}y - \frac{2}{3}(xy)^{-1/3}y$$

$$\frac{\partial}{\partial x}(xy) = y$$

$$\frac{\partial f}{\partial y} = \frac{3}{2}(xy)^{1/2}x - \frac{2}{3}(xy)^{-1/3}x$$

$$\frac{\partial}{\partial y}(xy) = x$$

$$10) f(x, y) = x^2 e^{-xy/2}$$

$$\frac{\partial f}{\partial x} = 2x e^{-xy/2} - x^2 \frac{y}{2} e^{-xy/2}$$

$$\frac{\partial f}{\partial y} = x^2 \cdot \frac{-x}{2} e^{-xy/2}$$

$$38) f(x, y, z) = y \tan(x^2 + z)$$

$$\frac{\partial f}{\partial x} = y \cdot \underline{2x} \sec^2(x^2 + z)$$

$$\frac{\partial}{\partial x}(x^2 + z) = 2x$$

$$\frac{\partial f}{\partial y} = \tan(x^2 + z)$$

$$\frac{\partial f}{\partial z} = y \sec^2(x^2 + z) \cdot 1$$

$$44) \quad g(s, t) = \ln(s^2 + 3st)$$

$$\frac{\partial^2 g}{\partial t^2} = \frac{\partial}{\partial t} \left(\frac{\partial g}{\partial t} \right)$$

$$\frac{\partial g}{\partial t} = \frac{1}{s^2 + 3st} \cdot 3s = \frac{3s}{s^2 + 3st}$$

$$\frac{\partial^2 g}{\partial t^2} = \frac{\partial}{\partial t} [3s(s^2 + 3st)^{-1}]$$

$$= 3s(-1)(s^2 + 3st)^{-2} \cdot 3s$$

$$= \frac{-9s}{(s^2 + 3st)^2}$$

$$42) f(x, y) = \sin(x - y)$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$$

$$\frac{\partial f}{\partial x} = \cos(x - y)$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \cos(x - y) = f_{xy}$$

$$= -\sin(x - y)(-1)$$

$$= \sin(x - y)$$